RES-4 closure: the STEP-5B layer-closure ratio is positive across the

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

first issued 2026-06-09 · this version issued 2026-06-09 · v1.1

1. Purpose and scope

The H-LAYER residual inventory (`hlayer-residual-inventory` v1.0) decomposes the distance between the amended-class conditional theorem and unconditional whole-Reading-H into six named items, of which three are independent open axes: RES-3 (unrestricted class, = DR-2; discharged for the crystallographic lattice class by R-026/R-027/R-028), RES-5 (matched-order to exact, = GAP-2; ESTIMATOR-UPGRADE CLOSED@controlled-error), and RES-4 (intensity interval). RES-4 was the one purely mechanical residual: the STEP-5B layer-closure margin chain was certified only at the three anchor intensities $I \in \{4 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$ (AddE floors $\rho = \times 59.4/\times 8.8/\times 2.6$), with the in-between interval left as a registered candidate follow-up script. This note closes RES-4 by an interval certification of $\rho(I)$ over the full operating-point interval. The scope is the anchor layer comparison ($\mu^2 = 5 \times 10^{-3}$), second-cumulant order, lattice competitor class – exactly the scope on which the chain is defined; nothing here touches RES-3 or RES-5.

2. The closure ratio as a function of intensity

At the anchor μ^2 the gap solution r_R, the dressed coupling $\lambda' = 3u + 30vM_R$, and the variance M_R are intensity-independent. The STEP-5B layer-closure ratio is

$$\rho(I) \;=\; \frac{K_{\rm b}(I)}{K(I)}, \qquad K_{\rm b}(I) = \frac{4\,(1 - a_0(I))\,m}{(\lambda' I)^2\,J_{\rm eff}(I)}, \qquad K(I) = 8 + 4\sqrt{14}\,\sqrt{n_{\rm pack}(I)},$$

with $\hat{r}(I) = r_R + 2\lambda' I$, $a_0(I) = 2\lambda' I/\hat{r}(I)$, $\theta_{\min}(I) = \sqrt{\hat{r}(I)}/(2q_0^2\sqrt{C})$, $n_{\rm pack}(I) = 16/\theta_{\min}^2$, $t_{\min}(I) = 2q_0 \sin(\theta_{\min}/2)$, $J_{\rm eff}(I) = J(t_{\min}, \hat{r})$, and $m = 4.32 \times 10^{-3}$ the Prop-A bulk layer margin (an INPUT here, intensity-independent by construction; its μ^2-robustness is ROBUSTNESS-MU2, separately closed). Every factor is a smooth monotone function of I: $\hat{r}$ and a_0 increase, $n_{\rm pack}$ decreases, and $K_{\rm b} \propto (1 - a_0)/(I^2 J_{\rm eff})$ decreases, so $\rho(I)$ is smooth and monotone decreasing on the interval.

3. Interval certification (RES-4 closed)

A 201-node sweep over $I \in [4 \times 10^{-4}, 2 \times 10^{-3}]$ at the anchor μ^2 establishes, with self-test asserts:

- **Anchor reproduction.** $\rho(4 \times 10^{-4}) = 59.4$, $\rho(10^{-3}) = 9.8$, $\rho(2 \times 10^{-3}) = 2.58$, matching the AddE floors $\times 59.4/\times 8.8/\times 2.6$ (the mid anchor differs by $\sim 11\%$ from the recorded 8.8 at this J-quadrature resolution; the binding endpoint reproduces $\times 2.6$ to 0.8%).
- **Positivity.** $\rho(I) > 1$ at every node; the grid minimum $\times 2.578$ occurs at the endpoint.

- **Derivative-sign certificate.** $\rho'(I) < 0$ at all 201 nodes ($\max_I \rho'(I) = -2.5\times10^3 < 0$), so ρ is strictly monotone decreasing on the continuum and the interval minimum is the ENDPOINT value.
- **Endpoint margin.** $\min_{[4\times10^{-4},2\times10^{-3}]} \rho = \rho(2\times10^{-3}) = \times2.578 > 1$, a margin of $\times1.58$ above unity.
- **Grid convergence.** The minimum is identical (to 0.000%) at $N = 200$ and $N = 100$, so no sub-grid structure is missed.

$$\boxed{\rho(I) \geq \rho(2\times10^{-3}) = \times2.58 > 1 \quad \text{for all } I \in [4\times10^{-4}, 2\times10^{-3}].}$$

The three-anchor caveat of RES-4 is removed: the STEP-5B layer closure holds on the full operating-point intensity interval, not merely at the sampled points.

4. Impact: one residual axis discharged, no tier flip

With RES-4 closed, the operating-point H-LAYER residual is covered on all three named axes at controlled-error grade: RES-3 (lattice class, R-026/R-027/R-028), RES-4 (this note), RES-5 (ESTIMATOR-UPGRADE controlled-error). This does NOT discharge H-LAYER and does NOT flip B1's tier. H-LAYER's analytic core – the isotropic dressing as the diagonal-Gaussian infimum and its beyond-diagonal / exact-free-energy content (RES-1 merged with RES-5, the GAP-2 frontier) – remains the deep obstruction, and the unrestricted-class axis (RES-3 for arbitrary real-point Q) remains open off the critical path. B1 stays T6 CONDITIONAL on {H-LAYER}; this is an increment that narrows the residual ledger, not a promotion.

5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"A finite grid does not prove monotonicity between nodes."* VALID-with-mitigation: the certificate is the derivative-sign test $\rho'(I) < 0$ at all 201 nodes together with grid convergence (N=200 vs N=100 identical minimum) and the smoothness of ρ (a composition of smooth monotone elementary functions of I). This is a controlled-error certification at the ROBUSTNESS-MU2 standard, not formal interval arithmetic; the note claims exactly that grade. A formal validated-numerics enclosure is a registered optional hardening, not required for the controlled-error closure.

(β) *"$m = 4.32\times10^{-3}$ is hardcoded; the closure could hide an intensity dependence of the bulk margin."* DISMISSED: m is the Prop-A bulk (layer) margin, which by construction is the second-order free-energy gap of the dressed layer and does not depend on the beyond-layer intensity I (the intensity enters only through K_b, K). The μ^2-dependence of m is the separate ROBUSTNESS-MU2 axis (CLOSED@[×0.5, ×2]). RES-4 is strictly the I-axis of the beyond-layer ratio at fixed anchor m.

(γ) *"Does RES-4 closure warrant a B1 promotion?"* DISMISSED: no. RES-4 is one of three independent residual axes; the deepest (RES-5 / GAP-2, exact free energy) and the full H-LAYER infimum are untouched. The note enacts no tier change.

Quantitative sanity check (mandatory): *magnitude* – the endpoint ×2.578 reproduces the AddE floor ×2.6 to 0.8%; *limit* – as $I \to 0$, $K_b \propto I^{-2} \to \infty$ so $\rho \to \infty$, consistent with $\rho(4\times10^{-4}) = 59.4$;

sign-direction – ρ decreases in I (the endpoint is the thinnest), matching the monotone ordering of the three anchors; *dimensional* – ρ is a dimensionless budget ratio; *reproducibility* – the minimum is grid-stable to 0.000% under doubling.

6. Operator review and acceptance (2026-06-09)

The operator reviewed and ACCEPTED this RES-4 closure (no B1 tier flip), with three adversarial points, each already reflected in the body and reaffirmed:

1. *Controlled-error, not formal interval arithmetic.* The closure rests on a derivative-sign grid certificate + grid convergence + the smoothness/monotonicity structure, so the safe ledger label is RES-4 : CLOSED@controlled-error; a T7 formal interval theorem would be an overclaim.

2. *Mid-anchor discrepancy recorded.* $\rho(10^{-3}) = 9.8$ differs from the recorded floor 8.8 by $\sim 11\%$ (J-quadrature resolution); the binding ENDPOINT $\rho(2 \times 10^{-3}) = 2.58$ is reproduced to 0.8%, so the verdict is unaffected. Now in the footer No-overclaim.

3. *m is an input.* $m = 4.32 \times 10^{-3}$ is the intensity-independent Prop-A bulk layer margin; RES-4 closes the I-axis residual only. The analytic origin of m (exact free energy, diagonal-Gaussian infimum) remains the H-LAYER core.

Accepted ledger statement (operator-recommended): at $\mu^2 = 5 \times 10^{-3}$,

$$\rho(I) = \frac{K_{\mathrm{b}}(I)}{K(I)} > 0 \text{ on } I \in [4 \times 10^{-4}, 2 \times 10^{-3}], \qquad \min_I \rho = \rho(2 \times 10^{-3}) = 2.578 > 1,$$

$$\text{RES-4}: \ \text{OPEN} \longrightarrow \text{CLOSED@controlled-error},$$

at fixed anchor μ^2, second-cumulant order, lattice competitor class; this does NOT discharge *H*-LAYER.

Resulting H-LAYER residual structure (operator): RES-3 closed for the lattice class; RES-4 closed@controlled-error interval; RES-5/GAP-2 the remaining deep frontier; RES-1/Prop-A the H-LAYER analytic core. **Decision: accept RES-4 closure; no B1 tier flip; next mainline** = RES-5/GAP-2 **and the Prop-A / H-LAYER analytic core.**

7. Result footer

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Result ID:              B5-BEYOND-LAYER-BOUND / H-LAYER RES-4 intensity-interval
                        closure
Precise statement:      The STEP-5B layer-closure ratio rho(I)=K_b(I)/K(I) is
                        strictly monotone decreasing (rho'(I)<0 at all 201
                        nodes) and rho(I) >= rho(2e-3) = x2.58 > 1 for all
                        I in [4e-4, 2e-3] at the anchor mu^2=5e-3; the three
                        AddE anchor floors x59.4/x8.8/x2.6 are reproduced. RES-4
                        (intensity interval) of the H-LAYER residual inventory
                        is CLOSED@interval (controlled-error).
Scope:                  Anchor mu^2, second cumulant, lattice competitor class;
                        controlled-error interval certification (derivative-sign
                        + grid convergence), NOT formal interval arithmetic.
Dependencies:           Math424-AddA (gap_solve, M_fast, U/V/Q0/C); Prop-A bulk
                        margin m=4.32e-3 (input); AddE anchor floors; hlayer-
                        residual-inventory v1.0 (RES-4 definition).
Evidence grade:         EXECUTED (hlayer_res4_intensity_sweep.py 6/6).
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Reproduction command:    python codes/vacuum/hlayer_res4_intensity_sweep.py
Expected output:         anchors x59.4/x9.8/x2.58; rho>1 on grid; rho'(I)<0 all
                         nodes; interval min = endpoint x2.578; 6/6 PASS.
Falsification gate:      A node with rho(I)<=1, or rho'(I)>=0 (non-monotone), or
                         a refinement revealing min < endpoint, in
                         I in [4e-4,2e-3] at the anchor mu^2.
Tier before / after:     No tier flip. B5 T5, B1 T6 on {H-LAYER} unchanged.
                         RES-4: three-point -> interval CLOSED (controlled-error).
No-overclaim statement:  RES-4 closure discharges the intensity-interval residual
                         ONLY. It does NOT discharge H-LAYER (RES-5/GAP-2 +
                         diagonal-Gaussian infimum remain) and does NOT promote
                         B1. Controlled-error grade, anchor mu^2, lattice class.
                         Mid-anchor rho(1e-3)=9.8 differs ~11% from the recorded
                         8.8 (J-quadrature resolution); the binding endpoint
                         rho(2e-3)=2.58 is reproduced to 0.8% (operator review
                         point 2), so the verdict is unaffected.
Next required action:    RES-5 (matched-order to exact / GAP-2) is the remaining
                         deep frontier; RES-1 (diagonal-Gaussian infimum, the
                         Prop-A analytic core) is the H-LAYER discharge target.
                         Optional: formal validated-numerics enclosure of rho(I).
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